DO NOW — RECALL

# Unfamiliar algorithm design and evaluation

Name: _____ Class: _____ Date: _____

**1. What is a precondition?**

**2. Why write edge-case tests?**

**3. What does a loop invariant express?**

**4. C# array [4,1,4,2]: compute sum at indexes 0 and 2, and number of distinct values.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Requirement for an algorithm to work correctly.

### 2. Model answer

To expose faults on boundaries/unusual inputs.

### 3. Model answer

A property that stays true after each loop iteration.

### 4. Model answer

Indexes 0 and 2 sum8; distinct values {4,1,2} count3.

### By the end, you can…

• State the precondition

• Trace the invariant

• Evaluate and test

### Vocabulary — know these terms

invariant — Condition maintained throughout a loop.

precondition — Assumption required for correctness.

boundary test — Tests around value/size limits.

time complexity — Growth in work as input increases.

NEW LEARNING

# Read, annotate and explain

### State the precondition

The input must be sorted; equal values then appear consecutively.

### Trace the invariant

Start with one distinct value for a nonempty array. Each change from previous value adds one.

### Evaluate and test

The algorithm scans once O(n) time, O(1) extra space; test empty, all equal and all distinct inputs.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace distinct count in [2,2,5,5,7,7,7].

### Predict first

Duplicates do not increment the counter because each equals the preceding element.

### Explain the method

Only changes to five and seven increment count; result is three distinct values.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Trace DistinctCount on [3,3,3,6,8,8]. Record count after each index. [6]

### Question 2

Describe a correct set of test cases for the algorithm and their expected outputs. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why the algorithm may fail on unsorted [2,1,2]. Give a repair and complexity consequence. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**The distinct-count algorithm is valid for unsorted inputs without modification.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.